① a) $\begin{bmatrix} 1 & \alpha & \gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix}$

$k = 14$ $\alpha = 0$ $\gamma = 0$

$\alpha^2 = 0 \rightarrow \alpha = 0$

$\beta = -1$

$\begin{bmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}$ $\begin{matrix} m = -2 \\ k = 14 \\ \alpha = 0 \\ \gamma = 0 \\ \beta = -1 \end{matrix}$

b) $\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 0 & 5 & -9\gamma & 1 \\ 0 & 0 & 3 & \alpha^2 \\ 0 & 0 & 0 & 2 \end{bmatrix}$

0

$\alpha = 0$ $\gamma = 0$

upper triangular matrix $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 0 & 1 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$

Mehmet Enes Celik

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③ $H = I - 2uu^T$ u is a unit vector so $\|u\| = 1$

a) $H^2 = I^2 - I \cdot 2uu^T - 2uu^T \cdot I + 4uu^T \cdot uu^T$

$= I(I - 2uu^T - 2uu^T) + 4u \cdot \|u\|^2 \cdot u^T$

$= I - 2uu^T - 2uu^T + 4u \cdot \|u\|^2 \cdot u^T$

As $\|u\|^2 = u^2 u$

$\|u\| = \pm\sqrt{u^T u}$ $\|u\| = 1$

So $H^2 = I - 4uu^T + 4uu^T$

$\boxed{H^2 = I}$

b) $HH^T = (I - 2uu^T)(I - 2uu^T)^T$ $[(A-B)^T = A^T - B^T]$

$= (I - 2uu^T)(I^T - (2uu^T)^T)$

$= (I - 2uu^T)(I - I(u^T)^T u^T)$ $[(AB)^T = B^T \cdot A^T]$

$= I - 2uu^T)(I - 2uu^T)$ $[(u^T)^T = u]$

$= I^2 - 2uu^T - 2uu^T + 4uu^T \cdot uu^T$

$= I - 4uu^T + 4u\|u\|^2 u^T$ $[\|u\| = 1]$

$= I - 4uu^T + 4uu^T$

$= \boxed{H \cdot H^T = I}$ $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

H is orthonormal matrix

c) $Hu = (I - 2u^T) \cdot u = Iu - 2uu^T u$

$= u - 2u\|u\|^2$ [I is identity matrix

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On my honor I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code violation of these principles will lead to a zero grade and is subject to harsh discipline issues.

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